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From IID to SVC: Spatial Structure in Areal Data with R-INLA

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Outline

1. Introduction
    a. Bayesian Inference
    b. Why INLA?

2. **Spatial Models in R-INLA**

    a. Besag/BYM/BYM2
    b. SVCs Models

3. Conclusions

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Frequentist vs Bayesian Statistics

Intro

	Frequentist	Bayesian
Data	**Random** → repeated samples	**Fixed** → observed once
Parameters	**Fixed** unknown constants	**Random** variables
Uncertainty	From sampling distribution	From posterior distribution
Intervals	Confidence Intervals *(long-run frequency)*	Credible Intervals *(probability statements)*

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Bayes' Theorem and the Posterior Distribution

Bayesian Inference:

Update prior beliefs after seeing data ⟶ Result: **posterior distribution**

- **Prior** $p(\theta)$: what we believe before data
- **Likelihood** $p(Y|\theta)$: what observed data tells us
- **Posterior** $p(\theta|Y)$: updated belief

Bayes' Theorem:

$$p(\theta|Y) = \frac{p(Y|\theta)\ p(\theta)}{p(Y)}$$

Interpretation:

Posterior $\propto$ Likelihood $\times$ Prior

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INLA = *Integrated Nested Laplace Approximation*

- Alternative to MCMC (Monte Carlo Markov Chain)
- Uses approximations instead of simulation
- Designed for a specific class of models:
  Latent Gaussian Models (LGMs)

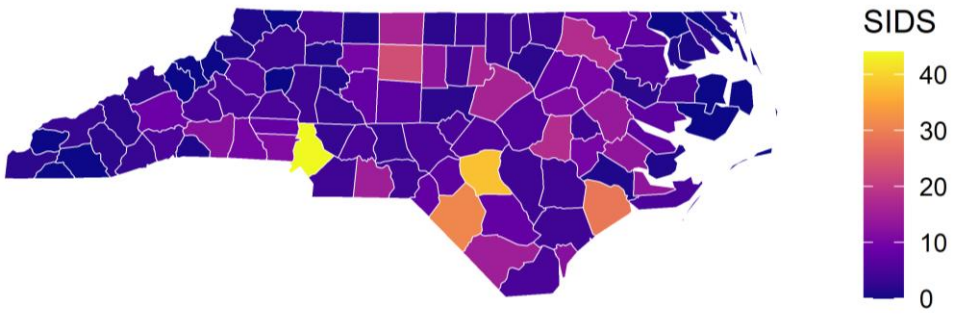
Why it's useful:

- Much **faster than MCMC**
- No convergence diagnostics
- Scales well for spatial data

INLA works especially well for spatial models because they are Gaussian and locally structured, which leads to sparse matrices and very efficient computations.

Example: North Carolina SIDS dataset

- Data about SIDS (Sudden Infant Death Syndrome) cases for 100 counties in North Carolina



$$y_i \sim Poisson(E_i * \theta_i)$$
$$\log(\theta_i) = \beta_0 + \beta_1 temp_i + \beta_2\ pm2.5_i + u_i$$

- y_i = observed SIDS cases in area i
- E_i = expected cases in area i (*number of births as a proxy for the population at risk*)
- $temp_i$ and $PM2.5_i$: two (**simulated**) enviromental variables in area i
- u_i = area-specific random effect

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Random Spatial Effects u_i

- Data from geographic areas are not independent
- Nearby areas tend to be similar

☞ We need to model spatial dependence explicitly

Introduce random effects to define the **type of spatial structure** in areal data:

1. No structure → **IID** (Independent and Identically Distributed)
2. Spatial structure → **Besag** model
3. Both → **BYM** / **BYM2** (Besag York Mollié) model

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Reference model (no random effects)

$$y_i \sim Poisson(E_i * \theta_i)$$
$$\log(\theta_i) = \beta_0 + \beta_1 temp_i + \beta_2\, pm2.5_i$$

```
m_base = inla(y ~ 1 + temp + pm25,
              family = "poisson",
              data = nc,
              E = E)
```

```
> summary(m_base)
Time used:
    Pre = 0.475, Running = 0.248, Post = 0.0273, Total = 0.75
Fixed effects:
              mean    sd 0.025quant 0.5quant 0.975quant   mode kld
(Intercept) -7.126 0.215     -7.547   -7.126     -6.705 -7.126   0
temp         0.035 0.009      0.018    0.035      0.052  0.035   0
pm25         0.008 0.020     -0.032    0.008      0.048  0.008   0
```


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Reference model (no random effects)

$$y_i \sim Poisson(E_i * \theta_i)$$
$$\log(\theta_i) = \beta_0 + \beta_1 temp_i + \beta_2\, pm2.5_i$$

```
m_base = inla(y ~ 1 + temp + pm25,
              family = "poisson",
              data = nc,
              E = E)
```

```
> summary(m_base)
Time used:
    Pre = 0.475, Running = 0.248, Post = 0.0273, Total = 0.75
Fixed effects:
              mean    sd 0.025quant 0.5quant 0.975quant   mode kld
(Intercept) -7.126 0.215     -7.547   -7.126     -6.705 -7.126   0
temp         0.035 0.009      0.018    0.035      0.052  0.035   0
pm25         0.008 0.020     -0.032    0.008      0.048  0.008   0
```

- The intercept represents the **baseline risk across all areas** on the **log scale** (because we're using a Poisson model with log link)
- On the original scale: exp(-7,126) ≈ 0.0008
  → this corresponds to a very small baseline rate, which is expected since SIDS is a rare event.

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Reference model (no random effects)

$$y_i \sim Poisson(E_i * \theta_i)$$
$$\log(\theta_i) = \beta_0 + \beta_1 temp_i + \beta_2\, pm2.5_i$$

```
m_base = inla(y ~ 1 + temp + pm25,
              family = "poisson",
              data = nc,
              E = E)
```

```
> summary(m_base)
Time used:
    Pre = 0.475, Running = 0.248, Post = 0.0273, Total = 0.75
Fixed effects:
              mean    sd 0.025quant 0.5quant 0.975quant   mode kld
(Intercept) -7.126 0.215     -7.547   -7.126     -6.705 -7.126   0
temp         0.035 0.009      0.018    0.035      0.052  0.035   0
pm25         0.008 0.020     -0.032    0.008      0.048  0.008   0
```

- exp(0,035) ≈ 1.036 → About **3.6% increase in risk per unit increase in temperature**
- 95% CI does NOT include 0 → **evidence of association**

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Reference model (no random effects)

$$y_i \sim Poisson(E_i * \theta_i)$$
$$\log(\theta_i) = \beta_0 + \beta_1 temp_i + \beta_2\, pm2.5_i$$

```
m_base = inla(y ~ 1 + temp + pm25,
              family = "poisson",
              data = nc,
              E = E)
```

```
> summary(m_base)
Time used:
    Pre = 0.475, Running = 0.248, Post = 0.0273, Total = 0.75
Fixed effects:
              mean    sd 0.025quant 0.5quant 0.975quant   mode kld
(Intercept) -7.126 0.215     -7.547   -7.126     -6.705 -7.126   0
temp         0.035 0.009      0.018    0.035      0.052  0.035   0
pm25         0.008 0.020     -0.032    0.008      0.048  0.008   0
```

- exp(0.008) ≈ 1.008 → About **0.8% increase per unit**, but…
- 95% CI includes 0 → **no clear evidence of effect**

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Residual Spatial Pattern

```r
# Pearson residuals
nc$fitted_base = m_base$summary.fitted.values$mean * nc$E
nc$pearson_base = (nc$y - nc$fitted_base) / sqrt(nc$fitted_base)

# Moran's I test on residuals
lw = nb2listw(nb, style = "W")
moran.test(nc$pearson_base, lw)
```

```
> moran.test(nc$pearson_base, lw)

        Moran I test under randomisation

data:  nc$pearson_base
weights: lw

Moran I statistic standard deviate = 3.9346, p-value = 4.166e-05
alternative hypothesis: greater
sample estimates:
Moran I statistic       Expectation          Variance
      0.241679909      -0.010101010       0.004094813
```

Interpretation:

- **significant positive Moran's I**: residuals are still **spatially clustered**
- near zero / non-significant little residual spatial dependence left

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IID model

Before moving to spatial random effects, we could try to fit the Poisson log-linear model plus an **unstructured area random effect**

$$y_i \sim Poisson(E_i * \theta_i)$$
$$\log(\theta_i) = \beta_0 + \beta_1 temp_i + \beta_2\, pm2.5_i + u_i$$
$$u_i \sim iid\ N(0, \sigma^2)$$

So the model is:

- adjusting for temperature and PM2.5
- allowing each area to have extra random heterogeneity
- but **not modelling spatial dependence** between neighbouring areas.

It may be useful to compare a **non-spatial random-effects model** against Besag/BYM/BYM2.

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IID model - Output

```
m_iid = inla(y ~ 1 + temp + pm25 +
        f(ID.area, model = "iid"),
        family = "poisson",
        data = nc,
        E = E)
```

```
> summary(m_iid)
Time used:
    Pre = 0.547, Running = 0.259, Post = 0.0468, Total = 0.853
Fixed effects:
              mean    sd 0.025quant 0.5quant 0.975quant   mode kld
(Intercept) -6.782 0.379     -7.539   -6.778     -6.050 -6.778   0
temp         0.018 0.015     -0.011    0.017      0.046  0.017   0
pm25         0.055 0.046     -0.035    0.055      0.145  0.055   0

Random effects:
  Name    Model
    ID.area IID model

Model hyperparameters:
                      mean   sd 0.025quant 0.5quant 0.975quant mode
Precision for ID.area 7.22 2.57       3.75     6.72      13.51 5.96
```


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IID model - Output

```
m_iid = inla(y ~ 1 + temp + pm25 +
        f(ID.area, model = "iid"),
        family = "poisson",
        data = nc,
        E = E)
```

Adding iid random effects:

- **reduces significance of temperature**
- **increases uncertainty overall**

```
> summary(m_iid)
Time used:
    Pre = 0.547, Running = 0.259, Post = 0.0468, Total = 0.853
Fixed effects:
              mean    sd 0.025quant 0.5quant 0.975quant   mode kld
(Intercept) -6.782 0.379     -7.539   -6.778     -6.050 -6.778   0
temp         0.018 0.015     -0.011    0.017      0.046  0.017   0
pm25         0.055 0.046     -0.035    0.055      0.145  0.055   0

Random effects:
  Name     Model
    ID.area IID model

Model hyperparameters:
                      mean   sd 0.025quant 0.5quant 0.975quant mode
Precision for ID.area 7.22 2.57       3.75     6.72      13.51 5.96
```


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IID model - Output

```
m_iid = inla(y ~ 1 + temp + pm25 +
        f(ID.area, model = "iid"),
        family = "poisson",
        data = nc,
        E = E)
```

```
> summary(m_iid)
Time used:
    Pre = 0.547, Running = 0.259, Post = 0.0468, Total = 0.853
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              mean    sd 0.025quant 0.5quant 0.975quant   mode kld
(Intercept) -6.782 0.379     -7.539   -6.778     -6.050 -6.778   0
temp         0.018 0.015     -0.011    0.017      0.046  0.017   0
pm25         0.055 0.046     -0.035    0.055      0.145  0.055   0

Random effects:
  Name     Model
    ID.area IID model

Model hyperparameters:
                      mean   sd 0.025quant 0.5quant 0.975quant mode
Precision for ID.area 7.22 2.57       3.75     6.72      13.51 5.96
```

1 / 7.22 ≈ 0.14

→ Moderate variability

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Besag Model

- Models **spatial dependence using neighbourhood structure**
- Each area depends only on its neighbours

$$u_i \mid u_{-i} \sim \mathcal{N}\left(\frac{1}{n_i}\sum_{j\sim i} u_j \ , \ \frac{1}{n_i\tau}\right)$$

where:

- $j \sim i$: neighbours of area i
- n_i :number of neighbours
- τ: precision parameter

1. The **expected value of** u_i is the **average of its neighbours**
2. The variance of u_i depends on the number of neighbors and on the precision parameter τ

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Besag Model - Output

```
m_besag = inla(y ~ 1 + temp + pm25 +
          f(ID.area, model = "besag", graph = nb),
          family = "poisson",
          data = nc,
          E = E)
```

```
> summary(m_besag)
Time used:
    Pre = 0.555, Running = 0.34, Post = 0.0485, Total = 0.944
Fixed effects:
              mean    sd 0.025quant 0.5quant 0.975quant   mode kld
(Intercept) -7.013 0.287     -7.578   -7.012     -6.449 -7.012   0
temp         0.026 0.012      0.003    0.026      0.049  0.026   0
pm25         0.044 0.033     -0.019    0.044      0.110  0.044   0

Random effects:
  Name    Model
    ID.area Besags ICAR model

Model hyperparameters:
                      mean   sd 0.025quant 0.5quant 0.975quant mode
Precision for ID.area 3.87 2.16       1.58     3.32       9.35 2.66
```


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Besag Model - Output

```
m_besag = inla(y ~ 1 + temp + pm25 +
          f(ID.area, model = "besag", graph = nb),
          family = "poisson",
          data = nc,
          E = E)
```

Reference model

```
Fixed effects:
              mean
(Intercept) -7.126
temp         0.035
pm25         0.008
```

```
> summary(m_besag)
Time used:
    Pre = 0.555, Running = 0.34, Post = 0.0485, Total = 0.944
Fixed effects:
              mean    sd 0.025quant 0.5quant 0.975quant   mode kld
(Intercept) -7.013 0.287     -7.578   -7.012     -6.449 -7.012   0
temp         0.026 0.012      0.003    0.026      0.049  0.026   0
pm25         0.044 0.033     -0.019    0.044      0.110  0.044   0

Random effects:
  Name    Model
    ID.area Besags ICAR model

Model hyperparameters:
                      mean   sd 0.025quant 0.5quant 0.975quant mode
Precision for ID.area 3.87 2.16       1.58     3.32       9.35 2.66
```


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Besag Model - Output

```
m_besag = inla(y ~ 1 + temp + pm25 +
          f(ID.area, model = "besag", graph = nb),
          family = "poisson",
          data = nc,
          E = E)
```

```
> summary(m_besag)
Time used:
    Pre = 0.555, Running = 0.34, Post = 0.0485, Total = 0.944
Fixed effects:
              mean    sd 0.025quant 0.5quant 0.975quant   mode kld
(Intercept) -7.013 0.287     -7.578   -7.012     -6.449 -7.012   0
temp         0.026 0.012      0.003    0.026      0.049  0.026   0
pm25         0.044 0.033     -0.019    0.044      0.110  0.044   0

Random effects:
  Name     Model
    ID.area Besags ICAR model

Model hyperparameters:
                      mean   sd 0.025quant 0.5quant 0.975quant mode
Precision for ID.area 3.87 2.16       1.58     3.32       9.35 2.66
```

Precision parameter τ

Variance: 1/3,87 ≈ 0.26
(moderate spatial variability)

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Residual Spatial Pattern – Besag model

```
nc$fitted_besag = m_besag$summary.fitted.values$mean * nc$E
nc$pearson_besag = (nc$y - nc$fitted_besag) / sqrt(nc$fitted_besag)
moran.test(nc$pearson_besag, lw)
```

```
> moran.test(nc$pearson_besag, lw)

        Moran I test under randomisation

data:  nc$pearson_besag
weights: lw

Moran I statistic standard deviate = -1.1408, p-value = 0.873
alternative hypothesis: greater
sample estimates:
Moran I statistic       Expectation          Variance
     -0.083721546      -0.010101010       0.004164552
```


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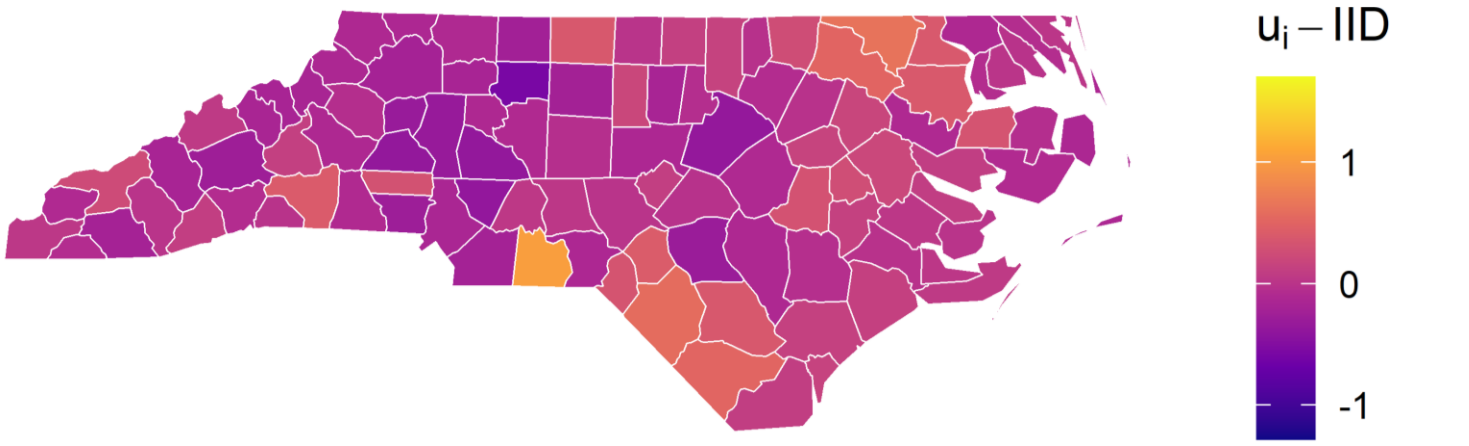
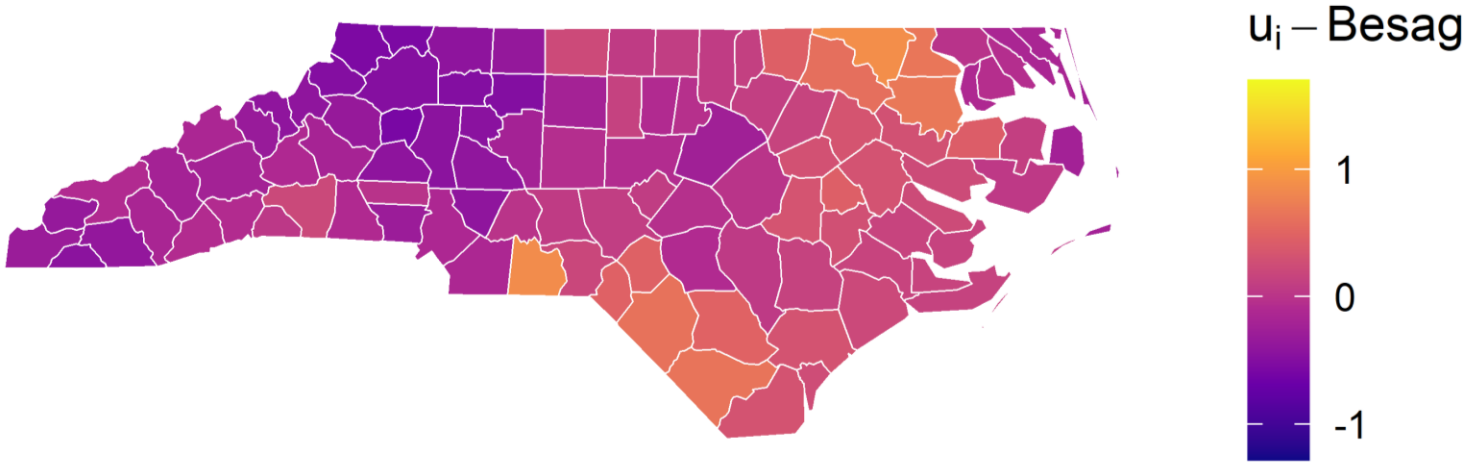
IID vs Besag Model - u_i





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What not to do

✗ *"The map of the spatial random effect u_i shows where PM 2.5 and Temperature have the largest effect."*

Not unless you explicitly fit a model with a **spatially varying coefficient**.
In the Besag model before, PM 2.5 has one global coefficient, and the spatial term is just residual area-level variation.

If we think the PM effect itself varies across space, we would need something like a **spatially varying coefficient** model, not just a spatial random intercept (we will see it later).

✓ *"After adjusting for temperature and PM2.5, there remains residual spatial structure given by u_i , indicating that not all spatial variation in SIDS cases is explained by the observed covariates."*

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BYM (Besag-York-Mollié) Model

Combines:

- **spatially structured effect** (Besag)
- **unstructured random noise** (iid)

$$u_i = v_i + w_i$$

where:

- $v_i \sim \text{Besag}$ $\rightarrow$ spatial effect $\rightarrow$ «**spatial clustering**»
- $w_i \sim \text{N}(0, \sigma^2)$ $\rightarrow$ iid noise $\rightarrow$ area-specific noise

1. Unlike the Besag model, not all variation is forced to be spatial
2. **Limitation:** Two variance components may be **hard to interpret**, there could be identifiability issues between spatial and iid parts

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BYM Model - Output

```
m_bym = inla(y ~ 1 + temp + pm25 +
        f(ID.area, model = "bym", graph = nb),
        family = "poisson",
        data = nc,
        E = E)
```

```
> summary(m_bym)
Time used:
    Pre = 0.643, Running = 0.413, Post = 0.101, Total = 1.16
Fixed effects:
              mean    sd 0.025quant 0.5quant 0.975quant   mode kld
(Intercept) -6.602 0.325     -7.246   -6.600     -5.969 -6.600   0
temp         0.010 0.013     -0.015    0.010      0.035  0.010   0
pm25         0.045 0.039     -0.031    0.045      0.122  0.045   0

Random effects:
  Name    Model
    ID.area BYM model

Model hyperparameters:
                                          mean      sd 0.025quant 0.5quant 0.975quant   mode
Precision for ID.area (iid component)  2177.60 2390.47     142.67  1426.58    8524.48 387.83
Precision for ID.area (spatial component)  2.72    1.03       1.28     2.53       5.28   2.18
```


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BYM Model - Output

```
m_bym = inla(y ~ 1 + temp + pm25 +
        f(ID.area, model = "bym", graph = nb),
        family = "poisson",
        data = nc,
        E = E)
```

```
> summary(m_bym)
Time used:
    Pre = 0.643, Running = 0.413, Post = 0.101, Total = 1.16
Fixed effects:
              mean    sd 0.025quant 0.5quant 0.975quant   mode kld
(Intercept) -6.602 0.325     -7.246   -6.600     -5.969 -6.600   0
temp         0.010 0.013     -0.015    0.010      0.035  0.010   0
pm25         0.045 0.039     -0.031    0.045      0.122  0.045   0

Random effects:
  Name    Model
    ID.area BYM model

Model hyperparameters:
                                          mean      sd 0.025quant 0.5quant 0.975quant   mode
Precision for ID.area (iid component)  2177.60 2390.47     142.67  1426.58    8524.48 387.83
Precision for ID.area (spatial component)  2.72    1.03       1.28     2.53       5.28   2.18
```


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BYM Model - Output

```
m_bym = inla(y ~ 1 + temp + pm25 +
        f(ID.area, model = "bym", graph = nb),
        family = "poisson",
        data = nc,
        E = E)
```

```
> summary(m_bym)
Time used:
    Pre = 0.643, Running = 0.413, Post = 0.101, Total = 1.16
Fixed effects:
              mean    sd 0.025quant 0.5quant 0.975quant   mode kld
(Intercept) -6.602 0.325     -7.246   -6.600     -5.969 -6.600   0
temp         0.010 0.013     -0.015    0.010      0.035  0.010   0
pm25         0.045 0.039     -0.031    0.045      0.122  0.045   0

Random effects:
  Name    Model
    ID.area BYM model

Model hyperparameters:
                                          mean      sd 0.025quant 0.5quant 0.975quant   mode
Precision for ID.area (iid component)  2177.60 2390.47     142.67  1426.58    8524.48 387.83
Precision for ID.area (spatial component)  2.72    1.03       1.28     2.53       5.28   2.18
```

Variance iid component: 1/2178 ≈ 0.00046 → **almost no unstructured noise**

Variance spatial component: 1 / 2.72 ≈ 0.37 → substantial **spatial variability**

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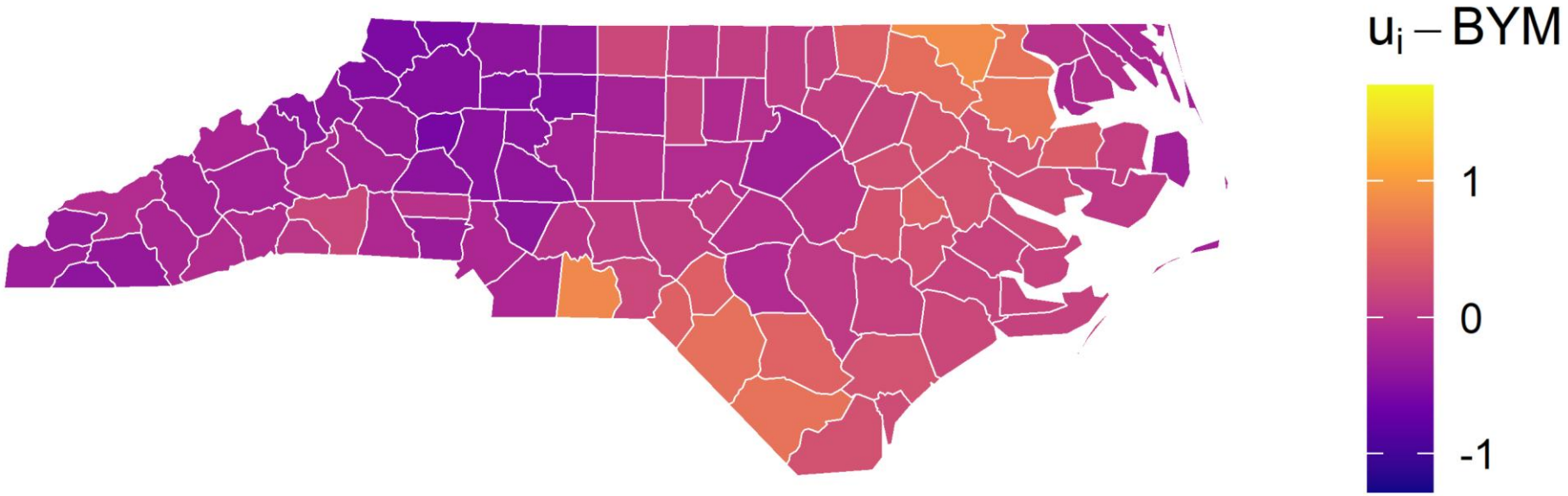
BYM Model - u_i



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BYM2 Model (Reparameterized BYM)

Reparameterization of BYM for better interpretation and computational efficiency

$$u_i = \frac{1}{\sqrt{\tau}}\left(\sqrt{\phi}\, v_i^* + \sqrt{1-\phi}\, w_i\right)$$

where:

- $v_i^* \rightarrow$ standardized spatial effect (variance = 1)
- $w_i \rightarrow$ iid effect
- $\tau \rightarrow$ total precision (**overall variability**)
- $\phi \in [0, 1] \rightarrow$ mixing parameter $\rightarrow$ **how much is spatial vs noise**

- $\phi \approx 0 \rightarrow$ mostly iid (no spatial structure)
- $\phi \approx 1 \rightarrow$ mostly spatial (strong clustering)

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BYM2 Model - Output

```
m_bym2 = inla(y ~ 1 + temp + pm25 +
          f(ID.area, model = "bym2", graph = nb, scale.model = TRUE),
          family = "poisson",
          data = nc,
          E = E)
```

```
> summary(m_bym2)
Time used:
    Pre = 11.3, Running = 0.406, Post = 0.062, Total = 11.7
Fixed effects:
              mean    sd 0.025quant 0.5quant 0.975quant   mode kld
(Intercept) -6.674 0.352     -7.376   -6.671     -5.992 -6.671   0
temp         0.014 0.014     -0.012    0.014      0.041  0.014   0
pm25         0.046 0.042     -0.036    0.046      0.130  0.046   0

Random effects:
  Name    Model
    ID.area BYM2 model

Model hyperparameters:
                      mean   sd 0.025quant 0.5quant 0.975quant  mode
Precision for ID.area 6.085 2.23      2.873    5.706      11.51 5.017
Phi for ID.area       0.588 0.23      0.141    0.608       0.95 0.778
```


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BYM2 Model - Output

```
m_bym2 = inla(y ~ 1 + temp + pm25 +
          f(ID.area, model = "bym2", graph = nb, scale.model = TRUE),
          family = "poisson",
          data = nc,
          E = E)
```

```
> summary(m_bym2)
Time used:
    Pre = 11.3, Running = 0.406, Post = 0.062, Total = 11.7
Fixed effects:
              mean    sd 0.025quant 0.5quant 0.975quant   mode kld
(Intercept) -6.674 0.352     -7.376   -6.671     -5.992 -6.671   0
temp         0.014 0.014     -0.012    0.014      0.041  0.014   0
pm25         0.046 0.042     -0.036    0.046      0.130  0.046   0

Random effects:
  Name     Model
    ID.area BYM2 model

Model hyperparameters:
                      mean   sd 0.025quant 0.5quant 0.975quant  mode
Precision for ID.area 6.085 2.23      2.873    5.706      11.51 5.017
Phi for ID.area       0.588 0.23      0.141    0.608       0.95 0.778
```

Overall variability across areas :
1 / 6.085 ≈ 0.16

→ **58,8%** of this variability is **spatial**

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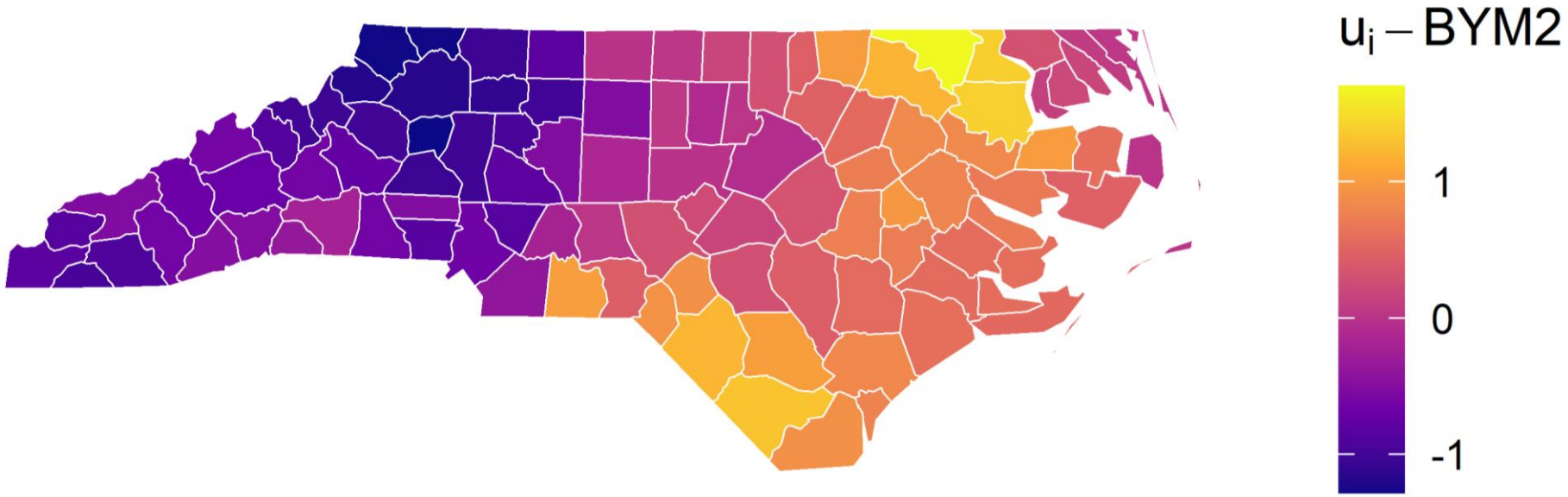
BYM2 Model - u_i



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SVC (Spatially-Varying Coefficient) Model

This model allows the **effects of temperature and PM2.5 to vary across space**, while also estimating an overall average effect.

$$y_i \sim Poisson(E_i * \theta_i)$$

$$\log(\theta_i) = \beta_0 + \beta_1 temp_i + \beta_2\ pm2.5_i + u_i temp_i + v_i\ pm2.5_i$$

$$= \beta_0 + [\beta_1 + v_{1,i}]\ temp_i + [\beta_2 + v_{2,i}]\ pm2.5_i$$

$$v_{1,i} \sim Besag$$

$$v_{2,i} \sim Besag$$

For each area and each covariate, we have

1. A **global effect** (β)
2. A **local deviation** from the global effect (v)

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SVC Besag Model - Output

```
m_svc_model = inla(y ~ 1 + temp + pm25 +
                   f(ID.area, temp, model = "besag", graph = nb) +
                   f(ID.area2, pm25, model = "besag", graph = nb),
                   family = "poisson",
                    data = nc,  E = E)
```

```
> summary(m_svc_besag)
Time used:
    Pre = 0.604, Running = 0.471, Post = 0.0651, Total = 1.14
Fixed effects:
              mean    sd 0.025quant 0.5quant 0.975quant   mode kld
(Intercept) -6.653 0.345     -7.335   -6.652     -5.979 -6.652   0
temp         0.012 0.014     -0.015    0.012      0.039  0.012   0
pm25         0.047 0.041     -0.032    0.047      0.127  0.047   0

Random effects:
  Name    Model
    ID.area Besags ICAR model
   ID.area2 Besags ICAR model

Model hyperparameters:
                           mean       sd 0.025quant 0.5quant 0.975quant    mode
Precision for ID.area   1643.68   599.55     795.61  1536.62    3121.18 1341.21
Precision for ID.area2 22213.89 24244.85    1513.45 14632.24   86590.45 4142.04
```


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SVC Besag Model - Output

```
m_svc_model = inla(y ~ 1 + temp + pm25 +
                   f(ID.area, temp, model = "besag", graph = nb) +
                   f(ID.area2, pm25, model = "besag", graph = nb),
                   family = "poisson",
                    data = nc,  E = E)
```

```
> summary(m_svc_besag)
Time used:
    Pre = 0.604, Running = 0.471, Post = 0.0651, Total = 1.14
Fixed effects:
              mean    sd 0.025quant 0.5quant 0.975quant   mode kld
(Intercept) -6.653 0.345     -7.335   -6.652     -5.979 -6.652   0
temp         0.012 0.014     -0.015    0.012      0.039  0.012   0
pm25         0.047 0.041     -0.032    0.047      0.127  0.047   0

Random effects:
  Name     Model
    ID.area Besags ICAR model
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Model hyperparameters:
                          mean       sd 0.025quant 0.5quant 0.975quant    mode
Precision for ID.area   1643.68   599.55     795.61  1536.62    3121.18 1341.21
Precision for ID.area2 22213.89 24244.85    1513.45 14632.24   86590.45 4142.04
```

- Temperature coefficient variability ≈ 0.0006
- PM2.5 coefficient variability ≈ 0.000045
- Very **little evidence that the effects vary across space**

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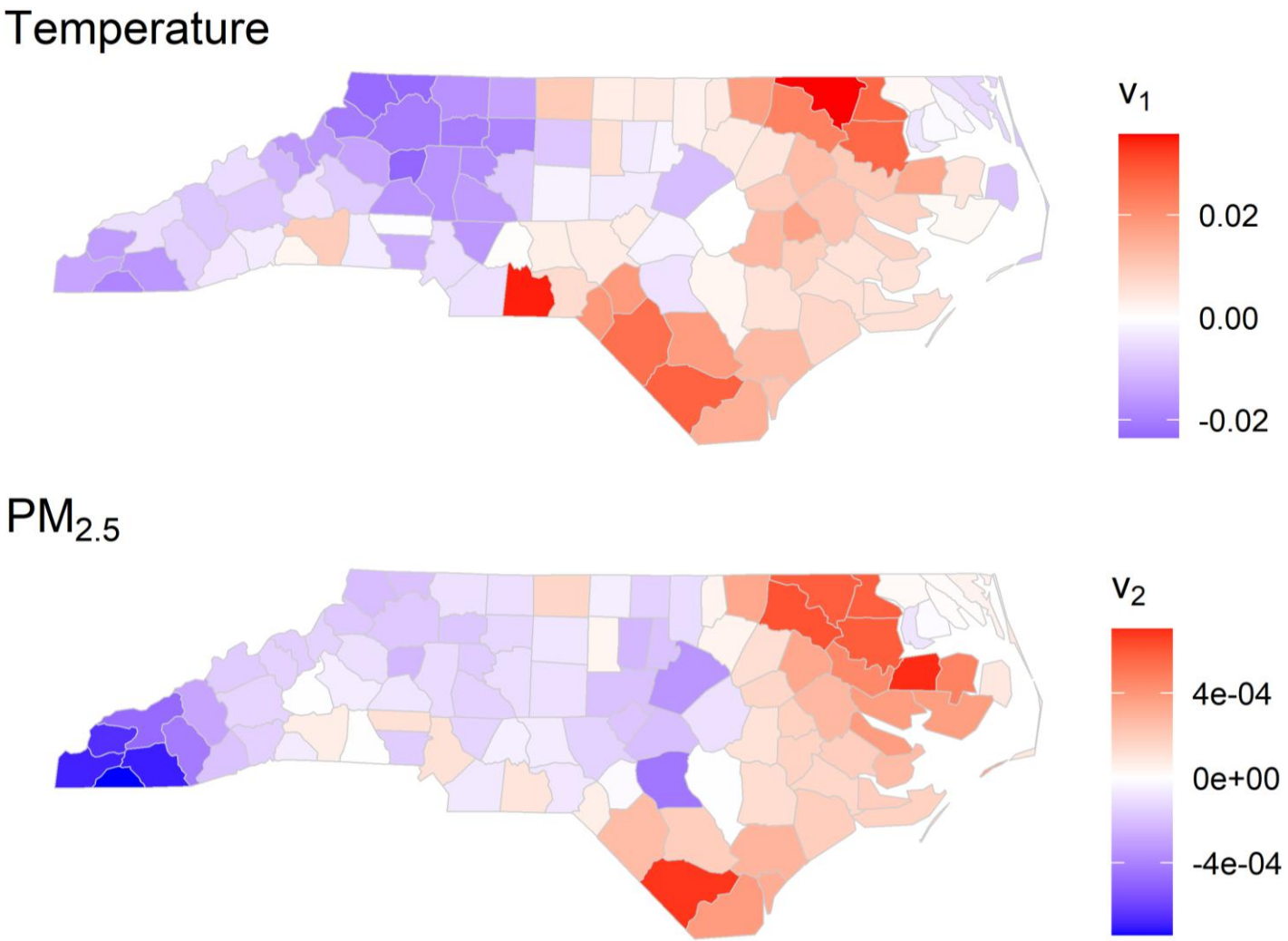
SVC Besag Model – v_1 e v_2



Model Comparison: Fit Criteria

Model	DIC	WAIC
Reference	493.68	499.19
iid	449.71	451.07
Besag	438.85	442.52
BYM	438.91	442.05
BYM2	441.95	441.97
SVC - Besag	440.58	443.02

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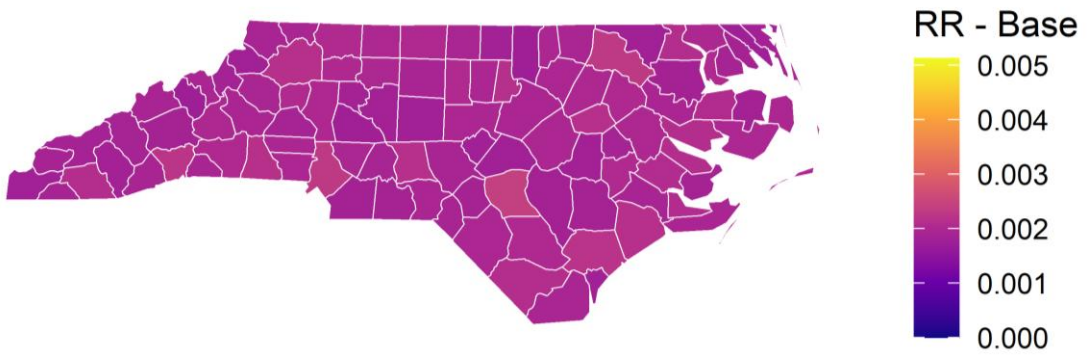
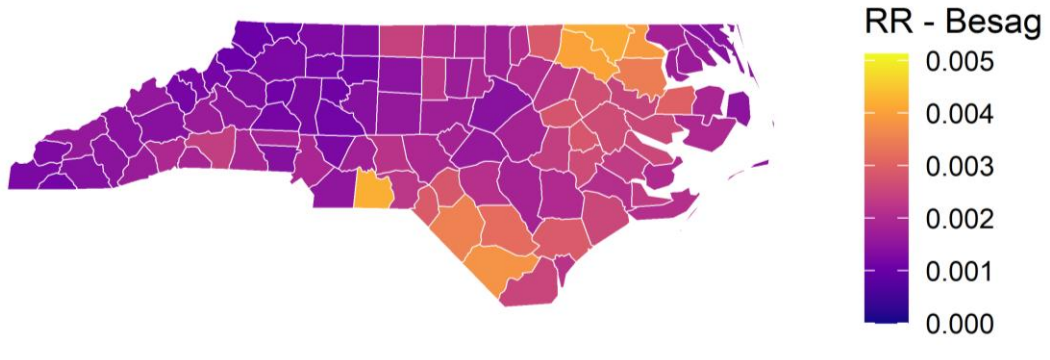
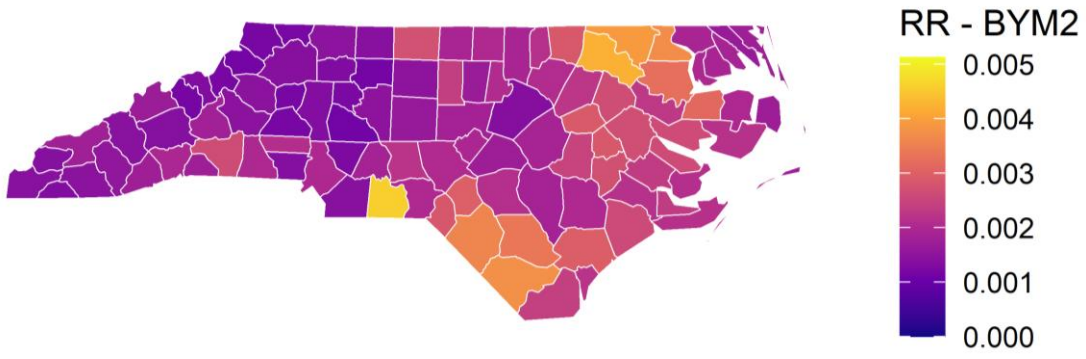
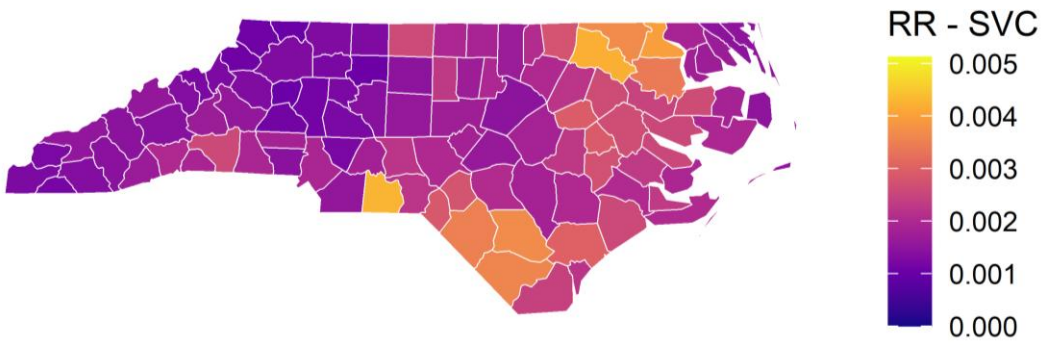
Model Comparison: Relative Risk









Conclusions

	Spatial Random Effect u_i	SVC v_i
Function	- Adjust for **spatial dependence in the outcome** - Capture **unexplained spatial variation**	- Allow covariate effects to **vary across space** - Model **spatial heterogeneity in relationships** between covariates and outcome
Affects	Baseline risk	Effect of covariate on risk
Interpretation	Where risk is higher or lower **after adjusting for covariates**	How the relationship between a covariate and the **risk changes across space**

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References

- (Blangiardo & Cameletti, 2015) - Spatial and spatio-temporal models with R-INLA. doi:10.1002/9781118950203
- (Lindgren & Rue, 2015) - Bayesian Spatial Modelling with R-INLA. doi:10.18637/jss.v063.i19
- (Bakka et al., 2018) - Spatial modeling with R-INLA: A review. doi:10.1002/wics.1443

Online Books:

- (Gómez-Rubio, 2021) - **Bayesian inference with INLA**
  https://becarioprecario.bitbucket.io/inla-gitbook/index.html
- (Moraga, 2023) - **Geospatial Health Data: Modeling and Visualization with R-INLA and Shiny**
  https://www.paulamoraga.com/book-geospatial/index.html

SVC Models:

- https://www.inla.r-inla-download.org/r-inla.org/doc/vignettes/svc.html
- https://groups.google.com/g/r-inla-discussion-group/c/X_g4NuqqHfY

R Code

https://github.com/marika-dagostini/inla_spatial_models_oviedo_26

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